Orthogonal Low-Rank Decomposition of Weight Matrix in Superposition

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1 Introduction

Superposition is a widely observed phenomenon in which neural networks represent more features than they have dimensions, a strategy that is viable when the underlying features are sufficiently sparse. This work investigates this phenomenon by analyzing a simple toy model that inherently lies in the superposition regime. Our primary objective is to decompose the model's weight matrix into a set of mutually independent components, each corresponding to a set of features.

2 Setup

An **encoder-decoder** model is used with **ReLU** activation in between.

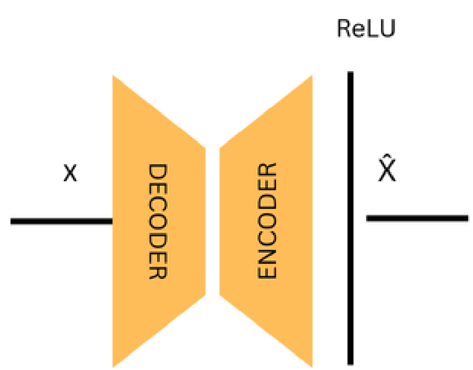

Figure 1: ToyModel

To simulate the sparsity required for superposition, we generate input data using a probabilistic masking approach. The data generation process is defined as:

$$x = r \odot m \tag{1}$$

Where $r \in \mathbb{R}^{n\times f}$ represents a matrix of random values sampled from a uniform distribution, and $m \in \{0,1\}^{n\times f}$ is a binary mask. The mask m is generated such that each element is 1 with probability $(1-s)$, where s denotes the target sparsity level. This ensures that, on average, a fraction s of the input features are set to zero, forcing the model to learn representations from a reduced-dimensional subset of the feature space.

3 Methodology

Given the encoder weight matrix W, we seek a decomposition of the form

$$W = \sum_{i=1}^{k} M_i,$$

where each W_i is a low-rank component capturing a distinct subset of features. An iterative approach is used to decompose W with the following loss function.

$$\begin{aligned}\mathcal{L} = \left\| W - \sum_{i=1}^{k} M_i \right\|_F^2 + \sum_{i=1}^{k} \sum_{j=i+1}^{k} \left\| |V_i| \odot |V_j| \right\|_F + \sum_{i=1}^{k} \sum_{j=i+1}^{k} \left\| |U_i| \odot |U_j| \right\|_F \\ + \sum_{r=1}^{n} \sum_{i=1}^{k} \sum_{j=i+1}^{k} \|M_i X_r\|_2 \, \|M_j X_r\|_2 \,.\end{aligned}$$

where V_i and U_i represent orthonormal bases for the column and row subspaces associated with M_i, respectively.

- **Weight Reconstruction Loss:** The first term ensures that the sum of the decomposed matrices accurately reconstructs the original weight matrix W.
- **Row-Space Separation Loss:** The second term encourages each component matrix to read from a distinct subset of input features, thereby promoting specialization among the decomposed components.
- **Column-Space Separation Loss:** The third term penalizes overlap between the output subspaces of different component matrices, reducing interference between their representations.
- **Competition Loss:** The fourth term encourages sparse component activations by penalizing simultaneous activations of multiple matrices for the same input, thereby promoting a winner-take-all behavior in which only one component is active at a time.

4 Results

We utilize input data consisting of $n = 50$ features and train the model to compress these features into a latent representation of dimension $d = 20$. The encoder weight matrix is therefore defined as $W \in \mathbb{R}^{20\times 50}$.

To decompose W, we employ $k = 5$ component matrices $\{M_1, \ldots, M_k\}$. Each component matrix M_i is constrained to be of rank 4 and is initialized using Xavier initialization prior to optimization.

The decomposition was performed by optimizing the proposed objective for 65 iterations. The recovered components closely matched the structure of the original weight matrix, yielding a high-quality decomposition and indicating that the proposed method effectively separates distinct feature representations.

Table 1: Summary of decomposition results. Lower values are better for reconstruction errors and subspace overlaps, while higher values are better for cosine similarities.

Metric	**Value**
Weight Reconstruction MSE	0.0038
Weight Reconstruction Cosine Similarity	0.9671
Activation Reconstruction MSE	0.0064
Activation Reconstruction Cosine Similarity	0.9587
Average Activation Norm	424.30
Average Column-Space Overlap	0.0033
Median Column-Space Overlap	0.0018
Average Row-Space Overlap	0.0016
Median Row-Space Overlap	0.0014

Table 2: Baseline reconstruction error and single-component ablations.

Configuration	**MSE Loss**	**% Increase**
No Ablation (Baseline)	0.006373	–
Ablate Component 1	0.023510	268.9%
Ablate Component 2	0.022581	254.3%
Ablate Component 3	0.023221	264.4%
Ablate Component 4	0.022924	259.7%
Ablate Component 5	0.023282	265.3%

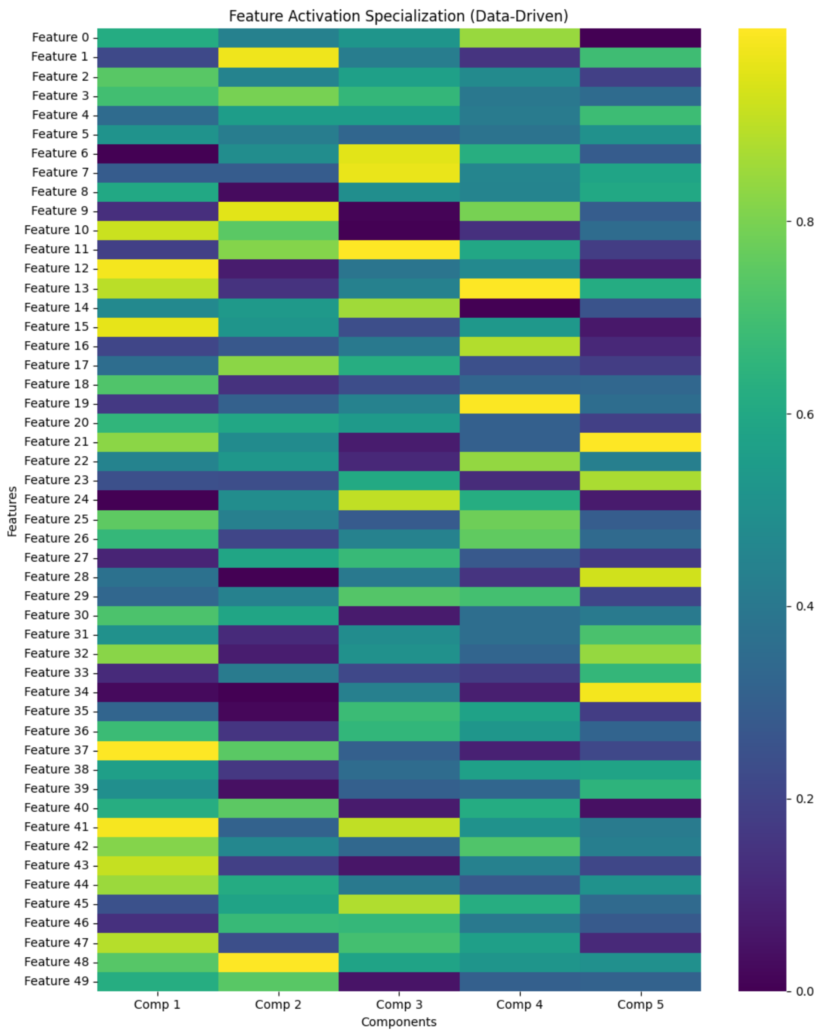

Figure 2: Component Specialisation

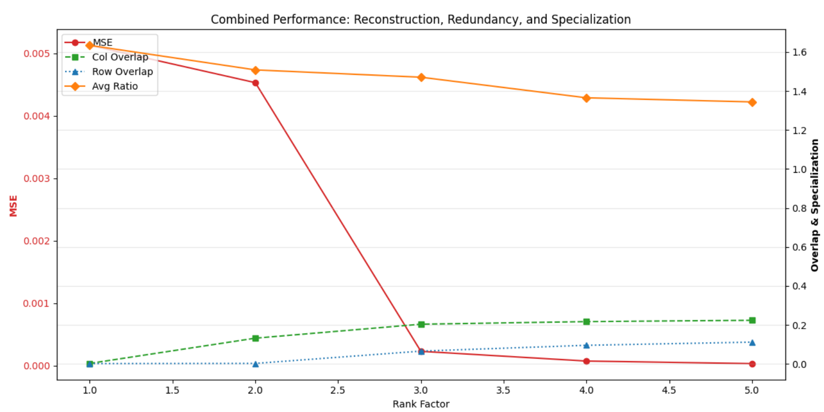

Figure 3: Metrics across different rank factors